THEORY OF COMPUTATION (CS302)

TOC Theoretical Viva Cheat Sheet

Frequently Asked Viva Questions

1. What is Theory of Computation?
2. Define DFA.
3. Define NFA.
4. Differentiate DFA and NFA.
5. What are epsilon transitions?
6. What is a finite automaton?
7. What are the components of DFA?
8. What are the components of NFA?
9. What is a Regular Language?
10. Explain the transition function.
11. What is a Regular Expression?
12. What is Kleene Star?
13. What is Kleene Plus?
14. State the Pumping Lemma.
15. Why is Pumping Lemma used?
16. What is a Context-Free Grammar?
17. What are terminals and non-terminals?
18. What is a parse tree?
19. Define ambiguous grammar.
20. What is Chomsky Normal Form?
21. Define Pushdown Automata.
22. What are stack operations?
23. What is acceptance by empty stack?
24. What is acceptance by final state?
25. State the equivalence of PDA and CFG.
26. Define Turing Machine.
27. What are the components of a Turing Machine?
28. What is the tape in a Turing Machine?
29. Define Decidability.
30. What is a Decidable Problem?
31. What is an Undecidable Problem?
32. Explain the Halting Problem.

33. What is Church-Turing Thesis?
34. Define Recursive Language.
35. Define Recursively Enumerable Language.
36. Differentiate Recursive and RE languages.
37. What is leftmost derivation?
38. What is rightmost derivation?
39. What is an Instantaneous Description (ID)?
40. What is CFG to PDA conversion?
41. What is PDA to CFG conversion?
42. What are applications of DFA?
43. What are applications of PDA?
44. What are applications of Turing Machines?
45. What is a production rule?
46. What is a derivation?
47. What is a language in TOC?
48. What is a state transition diagram?
49. What is a start state?
50. What is a final state?

Viva Preparation Tips

Focus on DFA, NFA, Regular Expressions, Pumping Lemma, CFG, CNF, PDA, Turing Machines, Decidability and the Halting Problem. These topics are commonly asked in viva examinations.